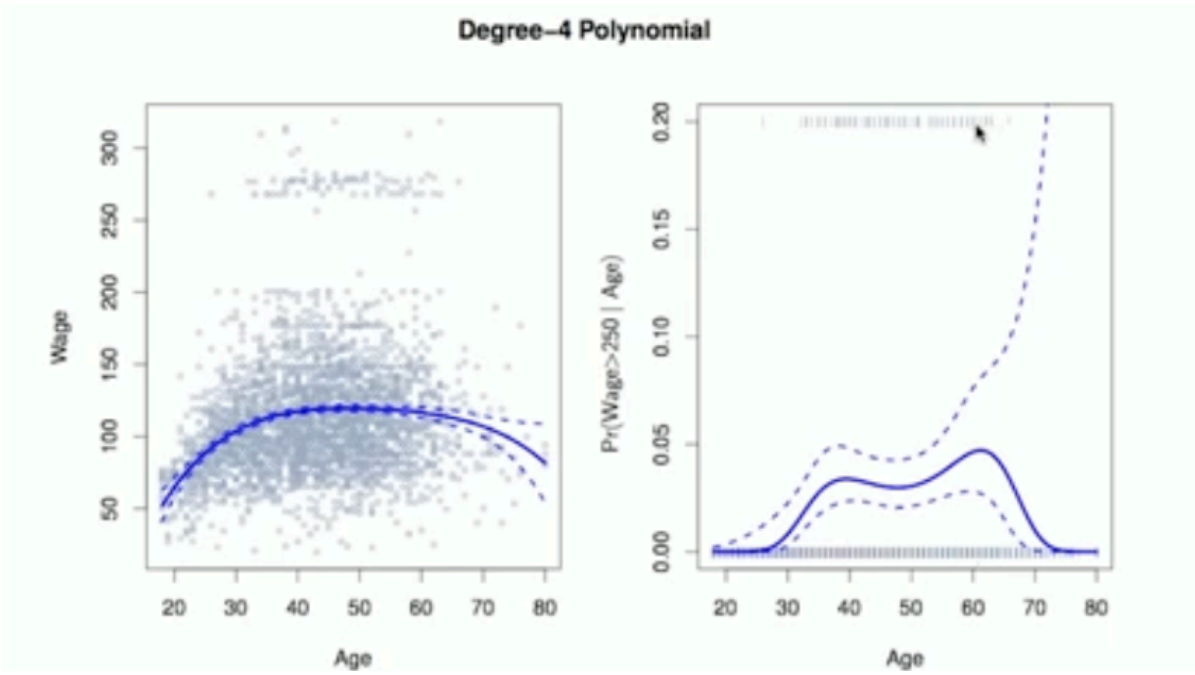
Polynomial Regression

- **Idea**: Even though the true relationship is non-linear, we still use linear regression but we add **higher-order terms** of the predictors and then we treat them as Multiple linear Rergession
- **Example**: Instead of $Y = \beta_0 + \beta_1 X$, we fit
  $Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + ... + \beta_d\ X\hat{}d$
- This is still a **linear model** in the parameters (β), so we can use ordinary least squares.
- **Pros**: Simple, captures curvature.
- **Cons**: High-degree polynomials become very **wiggly/unstable** at the end point (wiggly like a dog tail) , especially at the boundaries (Runge's phenomenon) which is bad for extrapolation, and they fit global behavior (one function for the entire range). One function affects all the parameter. (Changing a coefficient changes the entire function --> changing a data point on one side can affect the function in another side)
  - it is **not usually used** in **modern Machine Learning**
  - overfitting with high degree
  - hard to interpret
  - Numerically unstable
  - High degree without scaled feature will blowup the gradient calculation
    OxHadyy



left: Polynomial Regression
Right: we fit Polynomial to the Logistic Regression

Polynomial for Logistic Regression

$$\Pr(y_i > 250|x_i) = \frac{\exp(\beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \ldots + \beta_d x_i^d)}{1 + \exp(\beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \ldots + \beta_d x_i^d)}.$$

- to get Confidence interval, we compute the upper and lower bounds on the logit scale, and then invert it to get on probability scale